Transfer-learning classification results

CUB-200-2011 · ImageNet ResNet-18 · three repeated runs on 30 October 2026

Setup

All three logs use the same notebook settings: a pretrained ResNet-18 with a new 200-way layer, SGD at learning rate 0.001 and momentum 0.9, batch size 32, and seed 42. Training ran for 10 epochs. Six images per species were held out of the official training set for validation. The saved checkpoint is the epoch with the highest validation macro F1. In every run that was epoch 10, and the test set was scored once from that checkpoint.

Test results at the selected checkpoint

Scores are percentages. Macro precision, recall, and F1 treat each of the 200 species equally.

Run	Train acc.	Val acc.	Val F1	Test acc.	Test prec.	Test recall	Test F1
1	87.40	65.25	64.35	64.67	65.58	64.81	64.13
2	87.44	64.83	63.86	64.96	65.83	65.10	64.39
3	87.46	65.00	64.03	64.70	65.67	64.84	64.18
Mean	**87.44**	**65.03**	**64.08**	**64.78**	**65.69**	**64.92**	**64.23**

What the scores mean

A correct species is chosen for about 65 of every 100 test images. With 200 classes, guessing would be right about 0.5% of the time, so this is a real classifier, not a chance result. Macro precision, recall, and F1 stay within about one point of accuracy. The model is not succeeding only on a handful of species. The three test accuracies span 64.67% to 64.96%. Repeating the same run changes the score by less than a third of a point.

Training curves

The three curve plots have the same shape. Loss falls on both training and validation for all 10 epochs, from about 5.15 to 1.02 on the training set and from about 4.66 to 1.61 on validation. Around epoch 4 the training loss drops below the validation loss, and training accuracy pulls ahead, finishing near 87% while validation accuracy is still near 65%. That gap is overfitting: the network fits images it has already seen more tightly than new ones. Validation accuracy and macro F1 are still rising at epoch 10, so the run ended before the validation score leveled off. A longer run is the next check, not a change of architecture.

Files

runs/classification_resnet18_run1.csv, run2, and run3 hold one row per epoch plus a final test row. runs/curves_run1.png, curves_run2.png, and curves_run3.png are the matching loss and accuracy plots. Run 2 has the highest test accuracy and test macro F1. Run 1 has the highest validation macro F1. The difference is small enough that any of the three checkpoints tells the same story.

Checkpoint rule: highest validation macro F1. Test images were not used to choose the epoch.